

# 2022 Ph H2 Q1

Section: Our Dynamic Universe

Topic: Motion, Equations and Graphs —  
Projectiles

A football is kicked from the penalty spot (11 m from the crossbar) at  $17.0 \text{ m s}^{-1}$  and  $24.0^\circ$ . (a) Find the horizontal and vertical components of the initial velocity, (ii) show the travel time to the crossbar is 0.71 s, (iii) calculate the height  $h$  of impact. (b) If the same angle is used but with a smaller initial speed, state whether the ball hits/over/under the crossbar and justify.

## Worked solution

(a)(i) Components

Let  $u = 17.0 \text{ m s}^{-1}$ ,  $\theta = 24.0^\circ$ .

Horizontal component:

$$u_x = u \cos \theta = 17.0 \cos 24.0^\circ = 15.5 (\approx 15.53) \text{ m s}^{-1}.$$

Vertical component:

$$u_y = u \sin \theta = 17.0 \sin 24.0^\circ = 6.91 \text{ m s}^{-1}.$$

(a)(ii) Time to reach the crossbar — SHOW

Horizontal motion is uniform, so

$$t = d / u_x = 11.0 / 15.53 = 0.71 \text{ s}.$$

Therefore,  $t \approx 0.71 \text{ s}$  (shown).

(a)(iii) Height at impact

Vertical displacement after time  $t$ :

$$\begin{aligned}s &= u_y t - \frac{1}{2}gt^2 \\&= 6.91 \times 0.71 - 0.5 \times 9.8 \times (0.71)^2 \\&= 2.44 \text{ m.}\end{aligned}$$

Alternative method (at max height,  $v_y = 0$ ):

$$h = u_y^2 / (2g) = 2.44 \text{ m.}$$

(b) Smaller initial speed, same angle

Answer: The ball passes under the crossbar.

Justification: Smaller speed  $\Rightarrow$  smaller  $u_y$ , so maximum height  $u_y^2/(2g)$  is reduced; the ball cannot reach the same height as before.

### **Final answers**

**(a)(i)(A)  $u_x \approx 15.5$  ( $\approx 15.53$ )  $\text{m s}^{-1}$ .**

**(a)(i)(B)  $u_y \approx 6.91 \text{ m s}^{-1}$ .**

**(a)(ii)  $t = 0.71 \text{ s}$ .**

**(a)(iii)  $h \approx 2.44 \text{ m}$ .**

**(b) Passes under the crossbar.**

### **Revision tips**

- Resolve  $u$  into components:  $u_x = u \cos\theta$ ,  $u_y = u \sin\theta$ .
- Horizontal motion: constant velocity; vertical motion: acceleration  $-g$ .

- At maximum height:  $v_y = 0$ , so  $h_{\text{max}} = u_y^2 / (2g)$ .
- SHOW questions: keep more significant figures, round at the end.
- If speed reduces but angle is fixed, both range and height reduce.